Leander Paul

Senior Backend Engineer — Identity & Banking Platforms

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Summary

Senior backend engineer with 8+ years of experience across banking, identity and backend platforms. Building the mobile and online banking backend of ENBD X KSA, Emirates NBD's Saudi retail banking app, in a small AI-first squad: login, account opening, credit cards and payments, with production incident and disaster-recovery ownership. Previously worked on Zoho Accounts IAM.

Experience

Software Engineer — Sphere IT Consultants

Client: **Emirates NBD** · Dubai, UAE · Mar 2024 – Present

- Part of the team owning the mobile and online banking backend of ENBD X KSA, Emirates NBD's Saudi retail banking app: login, account opening, credit cards and payments.
- Built the mobile and online Request to Pay and instant-payment flow: balance check, payment verification and posting, orchestrating calls to other microservices.
- Architected the retail app's OTP, IVR and Nafath authorization microservice, with database-configured validation checks between steps.
- Built manual review for partially eligible credit-card applicants and integrated credit-scoring and employment-verification services into the application journey.
- Deliver in a small AI-first squad using Claude Code (AWS Bedrock) for coding, security review and debugging, with manual review of every change; squad velocity rose from 40–50 to 75–80 story points per sprint.
- Own escalated production incidents on live banking services, from root-cause analysis to shipped fix.
- Led disaster-recovery environment setup for the team's services, meeting SAMA requirements.

Technologies: Java, Spring Boot, Node.js, Express, Claude Code, AWS Bedrock

Full Stack Engineer (Contract) — WTV Global

London, UK · Mar 2023 – Sep 2023

- Helped design APIs and three RabbitMQ-connected NestJS microservices for Capture, a carbon-footprint platform, and developed backend services.
- Established end-to-end tests with database reset and reseeding before each run; services ran in Docker containers.

Software Engineer — Zoho Corporation

Chennai, India · Jan 2020 – Aug 2022

- Architected a hierarchical role-and-permission system for Zoho Accounts IAM and contributed to setting up SAML-based authorization.
- Built a proof of concept enabling LDAP queries against PostgreSQL through Apache Directory Server.
- Helped resolve Redis crashes caused by TTL-related CPU load by removing TTLs and balancing the load.
- Wrote migration scripts for sharded databases, set up and optimised an admin panel for inspecting and clearing user caches, and helped mentor junior engineers.

Technologies: Java, PostgreSQL, Redis

Full Stack Engineer & Team Lead — Phosphene AI
Chennai, India · Apr 2018 – Jan 2020

- Designed and built PixelView's image-upscaling pipeline using direct S3 uploads, Lambda and SQS, with on-demand compute that stopped after five idle minutes.
- Led three developers and built most of the React/Next.js frontend and Node.js backend, including Google integrations and Stripe payments.

Software Engineer (Freelance)
Chennai, India · Apr 2017 – Mar 2018

- Built and deployed two web applications on AWS EC2, including an e-commerce store with Razorpay payments and order/stock administration.

Projects

Shadow Platform — personal project; not deployed to production
github.com/shadow-library/platform · 2024 – Present

- Built an OIDC/OAuth 2.1 identity provider with PKCE, refresh-token rotation and reuse detection, RBAC, SAML 2.0, passkeys and TOTP.
- Implemented token-family and session revocation on refresh-token reuse, plus back-channel logout through a database outbox with retries and dead-lettering.
- AI: LangGraph workflows with a judge-and-repair loop, calling models through Synapse, my AI gateway with failover and budgets.
- Built solo with AI-assisted tooling; 10,000+ automated tests across unit, integration and end-to-end suites.

Open source: @shadow-library npm packages, maintained in the platform monorepo.

Skills

Backend: Java, Spring Boot, Node.js, TypeScript, Express, NestJS, Fastify, REST APIs, microservices

Identity & security: OAuth 2.0/2.1, OpenID Connect, SAML 2.0, WebAuthn/passkeys, TOTP MFA, RBAC, LDAP, OTP/IVR/Nafath authentication

Banking domain: retail banking, digital account opening, credit card applications, Request to Pay, instant payments

Data & messaging: PostgreSQL, Redis, MongoDB, RabbitMQ, AWS SQS

Cloud & delivery: AWS (EC2, S3, Lambda, SQS), Docker, Kubernetes (k3d), GitOps, CI/CD (GitHub Actions)

AI-assisted engineering: Claude Code, AWS Bedrock, AI-assisted code review, AI-assisted debugging

Education

MSc Computer Science (Distinction) — Queen Mary University of London — 2022 – 2023

B.Tech Computer Science — Anna University — 2016 – 2020